Relativity and Electromagnetism Homework Assignment 1

Submission deadline: 2024 March 21 (please submit by 17:00 CET)

[*Note:* This is the first homework assignment on the basics of electromagnetism. It contributes to 10% of the total points for final evaluation. You will be graded not only on the final result of a problem, but also on the reasoning and the intermediate steps. Please upload a (LaTeX / scanned) **pdf** copy of your answer sheet on Ufora by the deadline indicated above.]

Problem 1. The "Yukawa potential" is a screened Coulomb potential, given by,

$$V(\boldsymbol{r}) = -g^2 \frac{e^{-\alpha m r}}{r},$$

where g is a magnitude scaling constant, m the mass of the particle, and α another scaling constant. Find the electric field $\boldsymbol{E}(\boldsymbol{r})$, the charge density $\rho(\boldsymbol{r})$, and the total charge Q for the Yukawa potential above. **[1 point]**

Problem 2.

a. It is known that a spherical shell of radius R, carrying a uniform surface charge σ, spinning at a velocity ω about the z axis, produces a magnetic field described by the following vector potential:

$$\boldsymbol{A}(r,\theta,\phi) = \begin{cases} \frac{1}{3}\mu_0 R\omega\sigma\, r\, \sin\theta\, \widehat{\boldsymbol{\phi}}, & (r \le R), \\ \frac{1}{3}\mu_0 R^4\omega\sigma\, \frac{1}{r^2}\, \sin\theta\, \widehat{\boldsymbol{\phi}}, & (r \ge R). \end{cases}$$

Compute the magnetic field *inside* the shell, i.e., in the region $r \le R$.

b. Suppose a sphere of a superconducting material (below its critical temperature T_c) is placed in a uniform magnetic field $\boldsymbol{B} = B\,\widehat{\boldsymbol{z}}$. From an exercise done in the class, we know that the magnetic field inside a superconductor is zero and any current is confined to the surface. What is the induced surface current distribution? Try to sketch the (approximate) magnetic field lines. **[2 points]**

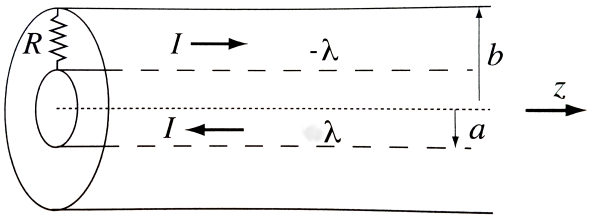
Problem 3. The Poynting theorem reads,

$$\frac{\partial}{\partial t}\left(u_{\text{mech}} + u_{\text{em}}\right) + \nabla\cdot\boldsymbol{S} = 0\,.$$

Here $u_{\text{em}} \equiv \frac{\epsilon_0}{2}E^2 + \frac{1}{2\mu_0}B^2$ is the energy density in the electric and magnetic fields, and u_{mech} is the mechanical (or thermal; essentially all non-electromagnetic) energy density. In class we defined the Poynting vector as $\boldsymbol{S} \equiv \frac{1}{\mu_0}\boldsymbol{E}\times\boldsymbol{B}$.

The Poynting vector $\boldsymbol{S}$ appears within a divergence in the Poynting theorem. Considering only the Poynting theorem, is the above definition of the Poynting vector unique, or is it possible to add a vector field to $\boldsymbol{S}$ such that it still satisfies the Poynting equation? What form must the additive vector field have? **[1 point]**

Problem 4. A cable is made from two infinite coaxial ideally conducting cylinders. The inner cylinder has radius a, charge per unit length λ, and carries a longitudinal current $\boldsymbol{I} = -I\,\hat{\boldsymbol{z}}$. The outer cylinder has radius $b > a$, charge per unit length $-\lambda$, and carries the current $\boldsymbol{I} = I\,\hat{\boldsymbol{z}}$ back in the opposite direction.



a. Obtain the electric and magnetic fields for this configuration.

b. Calculate the Poynting vector. In which direction does it point? Obtain its flux on a cross section of the cable.

c. If the $z < 0$ half of the cable is removed, and a resistor R is connected between the cylinders at $z = 0$, show that the resistor dissipates the power calculated in part b.

[2 points]

Problem 5. In class, we obtained the following expressions for the electric and magnetic fields for a point charge moving with a constant velocity $\boldsymbol{v}$. With $\boldsymbol{R} \equiv \boldsymbol{r} - \boldsymbol{v}t$ defined as the vector connecting the *present* position of the particle to the field point $\boldsymbol{r}$, and θ as the angle between $\boldsymbol{R}$ and $\boldsymbol{v}$,

$$\boldsymbol{E}(\boldsymbol{r},t) = \frac{q}{4\pi\epsilon_0}\frac{1-v^2/c^2}{\left(1-v^2\sin^2\theta/c^2\right)^{3/2}}\frac{\widehat{\boldsymbol{R}}}{R^2}\,,$$
$$\boldsymbol{B}(\boldsymbol{r},t) = \frac{1}{c^2}\,\boldsymbol{v}\times\boldsymbol{E}(\boldsymbol{r},t)\,.$$

Generalize the above expressions to the case of a line charge with charge per unit length λ moving with a constant velocity $\boldsymbol{v}$ along itself. You will need to integrate over θ. Express the magnetic field in terms of the current $I = \lambda v$. How do these electric and magnetic fields compare to the fields obtained in electrostatics and magnetostatics?

Please try to do the integration yourself rather than blindly copying from the solution manual. Depending on how you carry out the integration, you may find the following integral useful:

$$\int_{-1}^{1}\frac{dx}{(1-kx^2)^{3/2}} = \frac{2}{\sqrt{1-k}}$$

[2 points]

Problem 6. To the best of our knowledge, magnetic charges do not exist. This is the reason why $\nabla \cdot \boldsymbol{B} = 0$ (while $\nabla \cdot \boldsymbol{E}$ isn't). Imagine a universe that allows for a magnetic charge density ρ_m and a magnetic current density $\boldsymbol{J}_m$ governed by the continuity equation for the magnetic charges and currents: $\nabla \cdot \boldsymbol{J}_m + \partial \rho_m / \partial t = 0$. In such a universe, Maxwell's equations would have a more symmetric form,

$$\nabla \cdot \boldsymbol{E} = \frac{1}{\epsilon_0}\,\rho \qquad\qquad \nabla \times \boldsymbol{E} + \frac{\partial \boldsymbol{B}}{\partial t} = -\mu_0 \boldsymbol{J}_m$$

$$\nabla \cdot \boldsymbol{B} = \mu_0\,\rho_m \qquad\qquad \nabla \times \boldsymbol{B} - \mu_0 \epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} = \mu_0\,\boldsymbol{J}\,.$$

Do not worry about the fact that different constants sitting in front of the various terms – μ_0 and ϵ_0 can be made to disappear altogether by switching over to a different system of units.

Imagine a magnetic monopole q_m travelling at with a (large) constant velocity $\boldsymbol{v}$. What will the electric and magnetic field distributions be? Sketch the approximate electric and magnetic field lines for the magnetic monopole flying past.

Note: You do not need go through any nasty mathematics to come up with a solution for this problem. Feel free to use any expressions from class or from this assignment. **[2 points]**